

# Formulation Quizzes

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A company manufactures two lines of its product. It is planning production for a week in which it has available only 80 of a particular component needed for both lines. Inventory levels indicate that at most 50 of the first line should be made. The product lines require two hours and four hours of labor, respectively, and 280 hours of labor are available. If the contributions to profit of the two lines are \$60 and \$80, respectively, formulate the linear program to determine how many of each line to make to maximize their potential profit from the week's production.

$$\text{Max } 60P_1 + 80P_2$$

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$$P_1 \leq 50 \quad (\text{product 1})$$

$$P_1(2) + P_2(4) \leq 280 \quad (\text{time})$$

$$P_1 + P_2 \leq 80 \quad (\text{products made})$$

$$P_1, P_2 \geq 0$$

$$P_1, P_2 \text{ are integers}$$

### Formulation Exercise

A hardware store owner plans to diversify her business by renting out equipment as a sideline. Initially she plans to handle rug scrubbers, floor sanders, and chainsaws. A distribution estimates that respective monthly incomes from these items at \$150, \$100, and \$90. The distributor also advises that she should buy at least as many scrubbers as her total sanders and saws and no more than three saws. Scrubbers, sanders and saws together with the necessary supplies cost \$700, \$650, and \$500, respectively. In addition, she plans to buy a service contract on each item for an additional charge of \$25 each. She has \$9000 available. Her floor space is limited, and she can commit only 45 square feet of space to her rental operation. The three machines require 3, 2, and 2, feet of space, respectively for their storage. Her objective is to maximize income.

$$\text{Max } 150(R) + 100(F) + 90(C)$$

st

$$3(R) + 2(F) + 2(C) \leq 45 \quad (\text{space of inventory})$$

$$700(R) + 650(F) + 500(C) + 25(R+F+C) \leq 9000 \quad (\text{cost})$$

$$R, F, C \geq 0 \quad (\text{items qty})$$

$$R \geq F + C \quad (\text{as many rug scrubbers as floor sanders + chainsaws})$$

$$S \leq 3 \quad (\text{no more than three saws})$$